

Potential Outcomes

Causal Model: $Y_i = \alpha_0 + \alpha_1 D_i + u_i$, $Y_i(0) = \alpha_0$
 $\Rightarrow Y_i(0) = D_i Y_i(0) + (1-D_i) Y_i(0)$

ATE = $E[Y_i(1) - Y_i(0)] = \alpha_1$
ATT = $E[Y_i(1) - Y_i(0) | D_i=0]$

ATE with imperfect compliance is identified by N only if the treatment effect is constant (homogeneous) across individuals: $Y_i = Y_i(0) + Y_i(1) + A_i$

Causal Model w/ constant TE:
 $Y_i = Y_i(0) + D_i (Y_i(1) - Y_i(0)) = \alpha_0 + A_i D_i + u_i$

where $A_i \equiv Y_i(0) - Y_i(1)$, $A_i = E[Y_i(0)]$

$\hat{\beta}_0 = \frac{1}{N} \sum_{i=1}^N Y_i(0) - \frac{1}{N} \sum_{i=1}^N Y_i(1)$

$= \frac{1}{N} \sum_{i=1}^N (Y_i(0) - Y_i(1)) = \frac{1}{N} \sum_{i=1}^N Y_i(0) - \frac{1}{N} \sum_{i=1}^N Y_i(1)$

$= E[Y_i(0) | D_i=1] - E[Y_i(0) | D_i=0]$ by LIE

$= \alpha_1$ selection bias

3.1 Lec 7 (IV-LATE) - $\alpha_2 = D_i$

$Y_i, D_i \sim i$'s person's potential outcome when $D_i \in \{0,1\}$

$Z_i \sim$ instrument $\{1\}$ D

1) Random assignment of Z_i : $D_i(0), D_i(1)$ & Z_i

2) Exclusion Restriction: $Y_i(D, z) = Y_i(D)$

3) Monotonicity: $D_i(1) \geq D_i(0) \Rightarrow D_i(1) = 1$ if $D_i(0) = 0$

First-Stage: $D_i = \pi_0 + \pi_1 Z_i + \epsilon_i$

Reduced-Form: $Y_i = \delta_0 + \delta_1 Z_i + \eta_i$

	$Z_i = 0$	$Z_i = 1$
$D_i(0)$	$Y_i(D_i(0))$	$D_i(1)$
$D_i(1)$	$Y_i(D_i(0))$	$Y_i(D_i(1))$
AT	1	1
NT	0	0
Compliers	0	1
Defiers	1	0

g_i : denotes 4 groups; Defiers go away with monotonicity ($D_i(0) > D_i(1)$)

How to derive probabilities for population estimates of first stage coefficient π_1 :

$\pi_1 = E[D_i | Z_i=1] - E[D_i | Z_i=0]$

$E[D_i | Z_i=1] = E[D_i(0) | Z_i=1] = E[D_i(1)]$; by LIE we get:

$E[D_i | Z_i=1] = E[E[D_i | Z_i=1, g]]$

$= \sum_g \Pr(g) \cdot E[D_i | Z_i=1, g]$

$= \sum_g \Pr(g) \cdot E[D_i(0) | g]$ by random assignment independence $Z_i \perp (D_i(0), D_i(1))$ or $g^* \perp g$

Since $D_i(1) = 1$ for AT & Compliers:

$\Rightarrow E[D_i | Z_i=1] = \sum_g \Pr(g) \cdot E[D_i(0) | g] = \Pr(AT) + \Pr(Comp)$

Similarly since $D_i(0) = 0$ for AT & Defiers:

$\Rightarrow E[D_i | Z_i=0] = \sum_g \Pr(g) \cdot E[D_i(0) | g] = \Pr(AT) + \Pr(Defier)$

So our first stage coefficient (π_1):

$\pi_1 = E[D_i | Z_i=1] - E[D_i | Z_i=0] = \Pr(AT) + \Pr(Comp) - (\Pr(AT) + \Pr(Defier))$

$\Rightarrow \pi_1 = \Pr(Comp) - \Pr(Defier)$

$\Rightarrow$ If more compliers than defiers $\Rightarrow \pi_1 > 0$

$\Rightarrow \pi_1 = \Pr(Comp)$ if we have monotonicity (no defiers)

Head Start Program ~ IV-LATE framework

- 1) Relevance: $\pi_1 \neq 0$
- 2) Independence/Exclusion Restriction: $Z_i \perp (Y_i(0), Y_i(1))$ where Z_i only affects Y_i through D_i s.t. $E[Z_i | (Y_i(0), Y_i(1))] = 0$
- 3) Constant Treatment Effect (CATE): $Y_i = Y_i(0) + Y_i(1) = \alpha_1$ *

Causal Model: $Y_i = \alpha_0 + \alpha_1 D_i + u_i$

1st Stage: $D_i = \pi_0 + \pi_1 Z_i + \eta_i$ $\{ \pi_0 = E[D_i | Z_i=0] \}$

2nd Stage: Plug 1st stage into causal: $Y_i = \alpha_0 + \alpha_1 (\pi_0 + \pi_1 Z_i + \eta_i) + u_i$

$= \alpha_0 + \alpha_1 \pi_0 + \alpha_1 \pi_1 Z_i + \alpha_1 \eta_i + u_i$

Since $E[Z_i | (Y_i(0), Y_i(1))] = 0$ by random assignment/exclusion restriction we can estimate a reduced-form regression of Y_i on $(1, Z_i)$ to get

Reduced Form: $Y_i = \delta_0 + \delta_1 Z_i + \epsilon_i$

$\Rightarrow \hat{\beta}_1 = \frac{\delta_1}{\pi_1}$

$\hat{\pi}_1 = E[D_i | Z_i=1] - E[D_i | Z_i=0] = \frac{1}{N} \sum_{i=1}^N D_i - \frac{1}{N} \sum_{i=1}^N D_i = \bar{D}_1 - \bar{D}_0$

$\hat{\pi}_1 = \frac{1}{N} \sum_{i=1}^N D_i = \bar{D}_1$

Under (1-3*): $\frac{\delta_1}{\pi_1} \xrightarrow{p} \alpha_1 = ATE$

$\hat{\pi}_1 \sim$ represents compliance: the effect of HS offer on the actual enrollment in Head Start or compliance w/ the randomly assigned treatment Z_i (in IV-LATE, $\hat{\pi}_1$ ~ estimator of the fraction of compliers who would choose $D_i=0$ (not taking up the HS seat) if they were assigned to control group without an initial offer, and who would take up $D_i=1$ if they were assigned to the treatment group).

Under (1-2), if we have monotonicity ($D_i(1) \geq D_i(0) \Rightarrow$ no defiers)

$\Rightarrow \frac{\delta_1}{\pi_1} = E[Y_i(0) - Y_i(1) | Compliers] = LATE$

• (NT) Share of families who didn't take up HS seat despite being in treatment group $= (1 - 0.857) = 0.143 \sim E[D_i=0 | Z_i=1]$

• (AT) share of families who took HS seat despite being in control group $= \bar{D}_0 = \hat{\pi}_0 = 0.153$

• (Compliers) share of families who 'complied' w/ random assignment: π_1

* have no way of knowing which family is AT vs Complier

From reduced form ($Y_i = \delta_0 + \delta_1 Z_i + \epsilon_i$) $\Rightarrow \hat{\delta}_1 = \bar{Y}_1 - \bar{Y}_0$ where $\hat{\delta}_1 \sim$ intent to treat (ITT): the impact of HS offer (winning the lottery) on the outcome of kids. See below:

$N_1 = \sum_{i=1}^N Z_i$ (# units assigned to treatment) $\cdot \bar{D}_1 = \frac{1}{N_1} \sum_{i=1}^N D_i$ (avg. of D among lottery winners)

$N_0 = \sum_{i=1}^N (1-Z_i)$ (# units assigned to control) $\cdot \bar{Y}_1 = \frac{1}{N_1} \sum_{i=1}^N Y_i$ (avg. outcome of lottery winners)

$\hat{\delta}_0 = \bar{Y}_0 = \frac{1}{N_0} \sum_{i=1}^N Y_i \sim$ avg. outcome for losers in lottery (winners control group)

$\hat{\delta}_1 = \bar{Y}_1 - \bar{Y}_0 = (\frac{1}{N_1} \sum_{i=1}^N Y_i) - (\frac{1}{N_0} \sum_{i=1}^N Y_i)$

Second Stage (2SLS): $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 \hat{D}_i$

$\hat{\beta}_1 = \frac{\hat{\delta}_1}{\hat{\pi}_1} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{D}_1 - \bar{D}_0} = E[Y_i(1) - Y_i(0) | Compliers] = LATE \rightarrow \alpha_1$ CATE

• avg. causal effect of HS enrollment among compliers whose treatment (status) is induced by the instrument (lottery).

Smoking (Counseling IV) with Two Mutually-Exclusive Behaviors

Empirical story: Z_i (0,1) randomized counseling offer (Z_i treated).

- Outcome Y is latent behavior.
- Two binary mutually exclusive behaviors at delivery: $D_i = 1$ if the mother does not quit but changes behavior (e.g., low-intensity smoking / NRT only).
- $D_i = 0$ if the mother does quit and changes behavior (e.g., high-intensity smoking / NRT only).
- Potential outcomes $Y_i(D_i, Z_i)$ are any given Z_i and any given D_i (e.g., $Y_i(0,0)$ is the potential outcome if the mother does not quit and does not receive counseling).
- Random assignment: $(D_i(0), D_i(1), D_i(0), D_i(1), Y_i(0), Y_i(1)) \perp Z_i$ (no direct Z effect).
- Smoking given instrumentally exogenous for the uptake of treatment. For any uptake: $T(1) \geq T(0)$. Optionally, monotonicity $D_i(1) \geq D_i(0)$.

Principal strata for $(D_i(0), D_i(1)) = (D_i(1), D_i(0))$:

- NT: $(0,0) \rightarrow (0,0)$
- Always D1: $(0,0) \rightarrow (1,0)$
- Always D0: $(1,0) \rightarrow (1,0)$
- Defiers: $(1,0) \rightarrow (0,0)$
- Compliers: $(0,0) \rightarrow (0,1)$
- Always D0: $(1,0) \rightarrow (0,0)$
- Defiers: $(0,1) \rightarrow (0,0)$
- Compliers: $(0,0) \rightarrow (0,1)$

First-stages (what Z shifts).

$$\pi^{(1)} = E[D_1 | Z=1] - E[D_1 | Z=0] = \Pr(D_1=comp) - \Pr(D_1=def)$$

$$\pi^{(2)} = E[D_2 | Z=1] - E[D_2 | Z=0] = \Pr(D_2=comp) - \Pr(D_2=def)$$

$$\pi^{(T)} = E[T | Z=1] - E[T | Z=0] = \pi^{(1)} + \pi^{(2)}$$

Reduced form for birthweight.

$$\delta = E[Y | Z=1] - E[Y | Z=0] = E[Y(1,0) - Y(0,0) | D_1=comp] \Pr(D_1=comp) + E[Y(0,1) - Y(0,0) | D_2=comp] \Pr(D_2=comp)$$

$$- E[Y(1,0) - Y(0,0) | D_1=def] \Pr(D_1=def) - E[Y(0,1) - Y(0,0) | D_2=def] \Pr(D_2=def)$$

Which Wald ratio equals which LATE? (State the extra condition you use.)

$$\frac{\delta}{\pi^{(1)}} = \begin{cases} E[Y(1,0) - Y(0,0) | D_1=comp], & \text{if (i) } D_1\text{-monotonicity and (ii) either } \pi^{(2)} = 0 \text{ or } Y(0,1) = Y(0,0); \\ \text{signed mix of types,} & \text{otherwise.} \end{cases}$$

$$\frac{\delta}{\pi^{(2)}} = \begin{cases} E[Y(0,1) - Y(0,0) | D_2=comp], & \text{if (i) } D_2\text{-monotonicity and (ii) either } \pi^{(1)} = 0 \text{ or } Y(1,0) = Y(0,0); \\ \text{signed mix of types,} & \text{otherwise.} \end{cases}$$

$$\frac{\delta}{\pi^{(T)}} = \begin{cases} E[\tau_1 | D_1=comp] \Pr(D_1=comp) + E[\tau_2 | D_2=comp] \Pr(D_2=comp), & \text{if T-monotonicity;} \\ \Pr(D_1=comp) + \Pr(D_2=comp) & \text{otherwise,} \end{cases}$$

where $\tau_1 \equiv Y(1,0) - Y(0,0)$ is the birthweight gain from quitting, and $\tau_2 \equiv Y(0,1) - Y(0,0)$ is the gain from the non-quit behavior (e.g., low-intensity/NRT).

Fast "goofy" formulas to recover complier means (smoking context).

$$\frac{\delta Y \cdot D_1}{\pi^{(1)}} = E[Y(1,0) | D_1=comp], \quad \frac{\delta Y \cdot (1-D_1-D_2)}{\pi^{(1)}} = E[Y(0,0) | D_1=comp],$$

$$\frac{\delta X \cdot D_1}{\pi^{(1)}} = E[X | D_1=comp], \quad \frac{\delta X \cdot D_2}{\pi^{(2)}} = E[X | D_2=comp],$$

for any covariate X (e.g., maternal age). Analogous formulas hold swapping D_1/D_2 .

1. Draw the table: rows = NT / always-D1 / always-D2 / D1-comp / D2-comp (and defiers if allowed); columns Z=0,1; fill (D_1, D_2) and realized Y .
2. Compute $\pi^{(1)}, \pi^{(2)}, \pi^{(T)}$ and interpret (net shares moved by counseling).
3. Write the reduced form δ by differentiating the two Z columns.
4. Pick the relevant Wald ratio and state the extra condition that kills the "other-behavior" term.
5. (Optional) Use goofy products to back out $E[Y(1,0)]$, $E[Y(0,0)]$ or complier characteristics.

2) Derive an expression for δ_1 in the reduced form:

$\delta_1 = E[Y_i | Z_i=1] - E[Y_i | Z_i=0]$

By exclusion restriction, conditional on Z_i , realized $D_i = D_i(Z_i)$ and realized outcome $Y_i = Y_i(D_i(Z_i))$. Using LIE:

$\star E[Y_i | Z_i=0] = E_g[E[Y_i | Z_i=0, g]] = E_g[E[Y_i(D_i(0)) | Z_i=0, g]]$

$= E_g[E[Y_i(D_i(0)) | g]]$

Where $Z_i=0$ is dropped in the last equation b/c treatment is randomly assigned ($Z_i \perp (D_i(0), D_i(1), Y_i(0), Y_i(1)) \Rightarrow E_g[E[Y_i(D_i(0)) | g]] = \Pr(AT) \cdot E[Y_i(0) | AT]$

$+ \Pr(NT) \cdot E[Y_i(0) | NT] + \Pr(Comp) \cdot E[Y_i(0) | Comp] + \Pr(Defier) \cdot E[Y_i(0) | Defier]$

Similarly, conditional on being assigned to the treatment group $Z_i=1$

$\star E[Y_i | Z_i=1] = E_g[E[Y_i(D_i(1)) | g]] = \Pr(AT) \cdot E[Y_i(1) | AT] + \Pr(NT) \cdot E[Y_i(1) | NT]$

$+ \Pr(Comp) \cdot E[Y_i(1) | Comp] + \Pr(Defier) \cdot E[Y_i(1) | Defier]$

AT & NT have $Y_i(D_i(0)) = Y_i(D_i(1))$ as $D_i(0) = D_i(1)$, so the reduced form coefficient

$\delta_1 = E[Y_i | Z_i=1] - E[Y_i | Z_i=0]$

$\Rightarrow \delta_1 = \Pr(Comp) \cdot E[Y_i(1) - Y_i(0) | Comp] + \Pr(Defier) \cdot E[Y_i(1) - Y_i(0) | Defier]$ *

3) What is $\frac{\delta_1}{\pi_1}$ when $E[Y_i(1) - Y_i(0) | Complier] = E[Y_i(1) - Y_i(0) | Defier]$?

$\Rightarrow \frac{\delta_1}{\pi_1} = \frac{\Pr(Comp) \cdot E[Y_i(1) - Y_i(0) | Comp] + \Pr(Defier) \cdot E[Y_i(1) - Y_i(0) | Defier]}{\Pr(Comp) - \Pr(Defier)} = E[Y_i(1) - Y_i(0) | Comp]$

That is, the IV estimator would again give us the LATE of compliers!